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BACKUP POWER DEVICE AND PHOTOVOLTAIC SYSTEM

Abstract

A backup power device includes a power board and a regulation board. The regulation board is perpendicularly disposed on a surface of the power board. The power board includes a conductive wire and a switching component. The regulation board includes a semi-closed Rogowski coil and a regulation circuit. The semi-closed Rogowski coil crosses the conductive wire, and is configured to: induce a current in the conductive wire and generate an induction signal. The regulation circuit is configured to performing operation processing on the induction signal for output. The power board assembly further includes a controller, where the controller is connected to the regulation circuit and the switching component, and is configured to control on or off of the switching component based on an output signal of the regulation circuit.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims priority to Chinese Patent Application No. 202410272503.8, filed on Mar. 11, 2024, which is hereby incorporated by reference in its entirety.

TECHNICAL FIELD

[0002] The embodiments relate to the field of energy technologies, and to a backup power device and a photovoltaic system.

BACKGROUND

[0003] With continuous development of new energy technologies, a power board assembly used to implement power conversion is widely used in a plurality of different types of devices. For example, a photovoltaic system mainly includes a photovoltaic power generation device and a backup power device. The photovoltaic power generation device is configured to convert solar energy into a direct current, and the backup power device is connected between the photovoltaic power generation device and a power grid. The backup power device includes a power board assembly, and the backup power device is configured to convert, via the power board assembly, the direct current generated by the photovoltaic power generation device into an alternating current, and then transmit the alternating current to the power grid. During actual application, a current in the backup power device needs to be effectively detected, to ensure safety of the backup power device. In an existing backup power device, a sampling resistor or a current transformer is usually used to effectively detect the current. However, the foregoing two solutions have problems such as a large quantity of used components, a large occupied board area, and high costs. Therefore, it is difficult to widely apply the two solutions.

SUMMARY

[0004] The embodiments provide a cost-effective backup power device that has a good current detection function and occupies a small board area, and a photovoltaic system.

[0005] According to a first aspect, the embodiments provide a backup power device, including a power board and a regulation board, where the regulation board is disposed on a surface of the power board, and the regulation board is perpendicular to the power board; the power board includes a conductive wire and a switching component, and the switching component is connected to the conductive wire in series; the regulation board includes a semi-closed Rogowski coil and a regulation circuit, and the regulation circuit is connected to the semi-closed Rogowski coil; the semi-closed Rogowski coil crosses the conductive wire, and is configured to: induce a current in the conductive wire and generate an induction signal, and the regulation circuit is configured to generate a detection signal after performing operation processing on the induction signal; and the backup power device further includes a controller, where the controller is disposed on the power board or the regulation board, is connected to the regulation circuit and the switching component, and is configured to control on or off of the switching component based on the detection signal of the regulation circuit. In the backup power device provided in the embodiments, the semi-closed Rogowski coil has an advantage of low costs, which is conducive to wide application. The semi-closed Rogowski coil is used, so that the regulation board can be perpendicularly disposed on a side of the power board, and a good current detection function is implemented. In addition, the regulation board occupies a small board area in the power board, which helps implement a miniaturization design of the backup power device, so that the backup power device can be better used in a device having a high requirement on space.

[0006] It should be noted that the controller being configured to control on or off of the switching component based on the detection signal of the regulation circuit means that the controller further needs to refer to the detection signal when switching an on state or an off state of the switching component based on a control signal. Alternatively, it may be understood that, when the detection signal meets a condition, the controller may switch the on state or the off state of the switching component. When the detection signal does not meet the condition, the controller cannot switch the on state or the off state of the switching component.

[0007] For example, when the current in the conductive wire is less than a preset value, the regulation circuit generates a corresponding detection signal. In this case, the detection signal meets the condition, and the controller may switch the on state or the off state of the switching component.

[0008] Alternatively, when the current in the conductive wire is greater than or equal to the preset value, the regulation circuit generates a corresponding detection signal. In this case, the detection signal does not meet the condition, and the controller cannot switch the on state or the off state of the switching component.

[0009] Alternatively, it may be understood that, in some cases, the switching component does not have a turn-on/off capability under a large current; otherwise, a bad case like arcing may occur. Therefore, the switching component is allowed to switch between the on state and the off state only when the current in the conductive wire is small. Therefore, use safety of the switching component can be ensured.

[0010] During specific disposing, the semi-closed Rogowski coil may be in a semi-circular ring shape, and the conductive wire penetrates a center of the semi-closed Rogowski coil, so that the semi-closed Rogowski coil can effectively and accurately detect the current flowing through the conductive wire.

[0011] In an example, the regulation board includes a dielectric substrate, and the semi-closed Rogowski coil is located in the dielectric substrate. Alternatively, it may be understood that the semi-closed Rogowski coil may be packaged in the dielectric substrate, to ensure safety of the semi-closed Rogowski coil.

[0012] Alternatively, in an example, the semi-closed Rogowski coil is located on a side of the dielectric substrate. During preparing, the semi-closed Rogowski coil may be disposed on the side of the dielectric substrate through soldering or the like, facilitating preparation.

[0013] In an example, the semi-closed Rogowski coil is located between the regulation circuit and the conductive wire, so that the regulation circuit does not block between the semi-closed Rogowski coil and the conductive wire. In this way, detection effect of the semi-closed Rogowski coil is effectively ensured.

[0014] In an example, the semi-closed Rogowski coil includes a first end and a second end; and the semi-closed Rogowski coil is formed by a spiral wire, and both ends of the spiral wire are located at the first end or the second end. Alternatively, it may be understood that, signal output ends of the semi-closed Rogowski coil are all located at a same end of the semi-closed Rogowski coil, to effectively connect to the regulation circuit.

[0015] During specific disposing, in the semi-closed Rogowski coil, one end of the spiral wire is located at the first end, and the other end of the spiral wire extends from the second end along an axis of the semi-closed Rogowski coil to the first end; or one end of the spiral wire is located at the second end, and the other end of the spiral wire extends from the first end along the axis of the semi-closed Rogowski coil to the second end.

[0016] In an example, an edge of the regulation board includes a conductive protrusion, and the regulation circuit is electrically connected to the conductive protrusion; and the power board includes a conductive hole, the conductive protrusion is inserted into the conductive hole, and the conductive protrusion is electrically connected to the conductive hole. Through a connection between the conductive protrusion and the conductive hole, the power board and the regulation

board can be fastened, and a signal connection between the power board and the regulation board can be implemented.

[0017] During specific disposing, the regulation board includes two conductive protrusions, and the two conductive protrusions are both located at an edge of a same side of the regulation board; the power board includes two conductive holes, and the two conductive holes are connected to the two conductive protrusions in a one-to-one correspondence; and the two conductive holes are distributed on two sides of the conductive wire. By disposing the two conductive holes and the two conductive protrusions, connection stability and connection strength between the regulation board and the power board can be improved. The two conductive holes are distributed on the two sides of the conductive wire, so that after the power board is inserted into the regulation board, the semi-closed Rogowski coil can cross the conductive wire, facilitating mounting.

[0018] During specific disposing, the switching component is any one of a relay, an insulated gate bipolar transistor, and a metal-oxide-semiconductor field-effect transistor.

[0019] In an example, the backup power device may further include a housing, an input port, and an output port; and both the input port and the output port penetrate the housing, so that an external power-consuming device and an external power supply device are connected to the backup power device. Both the power board and the regulation board are located in the housing, and the conductive wire is connected between the input port and the output port. An external alternating current may be transmitted from the input port to the backup power device, and then output from the output port. In the backup power device provided in embodiments, a current detection function may be implemented through the foregoing power board and the regulation board, and use safety and reliability are good. In addition, the power board and the regulation board have advantages of miniaturization and low costs, so that the backup power device using the power board and the regulation board also has advantages of miniaturization and low costs, which facilitates wide application of the backup power device and good usage of the backup power device in a scenario having a high requirement on space.

[0020] In an example, the backup power device further includes a circuit breaker, the circuit breaker is connected between the conductive wire and the output port, and the circuit breaker is configured to connect or disconnect a conductive path between the conductive wire and the output port. The circuit breaker may be a device having a large current connection or disconnection capability. It may be understood that, in the backup power device, the switching component usually does not have the large current connection or disconnection capability. Therefore, in the backup power device provided in the embodiments, an additional circuit breaker is provided, so that when a working condition like a short circuit occurs in an external power-consuming device connected to the output port, a transmission path of a current may be cut off by using the circuit breaker. In this case, the switching component in the backup power device may keep in an on state.

[0021] For example, when the current in the conductive wire is greater than or equal to the preset value, the regulation circuit generates a corresponding detection signal. In this case, the detection signal does not meet the condition, and the controller cannot switch the on state or the off state of the switching component. Therefore, the controller may cut off the transmission path of the current by controlling the circuit breaker.

[0022] According to a second aspect, the embodiments further provide a photovoltaic system, including a photovoltaic power generation device, an inverter, and the foregoing backup power device. The inverter is connected between the photovoltaic power generation device and the backup power device, and the inverter is connected to the input port. The inverter is configured to: convert a direct current generated by the photovoltaic power generation device into an alternating current, and then transmit the alternating current to the input port. The output port is configured to connect to a power grid.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0023] FIG. **1** is a diagram of an application scenario of a power board assembly according to an embodiment;
[0024] FIG. **2** is a diagram of a structure of a conventional power board assembly according to an embodiment;
[0025] FIG. **3** is a diagram of a structure of a power board assembly according to an embodiment;
[0026] FIG. **4** is a circuit diagram of a regulation circuit according to an embodiment;
[0027] FIG. **5** is a diagram of an exploded structure of a power board assembly according to an embodiment;
[0028] FIG. **6** is a diagram of a side structure of a power board assembly according to an embodiment of this application;
[0029] FIG. **7** is a diagram of a structure of a semi-closed Rogowski coil according to an embodiment;
[0030] FIG. **8** is a diagram of a structure of a backup power device according to an embodiment;
and
[0031] FIG. **9** is a diagram of an application scenario of a backup power device according to an embodiment.

DETAILED DESCRIPTION OF EMBODIMENTS

[0032] To make the objectives, solutions, and advantages clearer, the following further describes the embodiments in detail with reference to the accompanying drawings.

[0033] To facilitate understanding of a backup power device provided in embodiments of the embodiments, the following first describes an application scenario of the backup power device.

[0034] The backup power device provided in embodiments may be used in a plurality of devices or scenarios in which transmission of electric energy needs to be connected or disconnected.

[0035] For example, as shown in FIG. **1**, in an example provided in the embodiments, the photovoltaic system may include a photovoltaic power generation device, an inverter, and a backup power device. The photovoltaic power generation device is configured to convert light energy into a direct current, and the inverter is configured to: convert the direct current generated by the photovoltaic power generation device into an alternating current and then transmit the alternating current to a power grid. The backup power device is connected between the inverter and the power grid, and is configured to implement connection or disconnection between the inverter and the power grid. In other words, the backup power device may connect a power supply path between the inverter and the power grid, so that electric energy generated by the photovoltaic power generation device can be transmitted to the power grid. Alternatively, the backup power device may disconnect the power supply path between the inverter and the power grid, so that the electric energy generated by the photovoltaic power generation device cannot be transmitted to the power grid.

[0036] As shown in FIG. **2**, the backup power device includes a power board assembly **01**. The power board assembly **01** may include a printed circuit board (PCB) **011**, and the printed circuit board includes a conductive wire **012**. The conductive wire **012** is configured to transmit an alternating current, so that the alternating current output from the inverter can be transmitted to the power grid through the conductive wire **012**. In addition, a switching component like a circuit breaker **013** is further disposed on the printed circuit board. The switching component **013** is connected in series to the conductive wire **012**, and is configured to connect or disconnect the conductive wire **012**, to implement an on or off function of the power board assembly **01**.

[0037] In an actual application, to ensure use safety of the power board assembly **01**, a current flowing through the conductive wire **012** needs to be effectively detected, to ensure use safety of the power board assembly **01**.

[0038] In the existing power board assembly **01**, two solutions are usually used to effectively detect the current in the conductive wire **012**. One solution is to use a sampling resistance, and the other solution is to use a current transformer (CT). In the sampling resistor solution, a sampling resistor, and components like an amplifier matching the sampling resistor, an isolation operational amplifier, and a comparator may be disposed on the printed circuit board **011**. In the sampling resistor solution, a large quantity of components is used, occupying a large board area. In addition, the isolation operational amplifier needs to be powered by three independent isolated power supplies. Thus, the sampling resistor solution has high costs and occupies a large board area, which is not conducive to wide application. For the current transformer solution, the current transformer needs to be installed in the outside, occupying large space and requiring a complex signal cable connection. In addition, the current transformer has high costs, and therefore is not conducive to wide application.

[0039] Therefore, embodiments provide a cost-effective power board assembly that has a good current detection function and occupies a small board area, and a backup power device equipped with the power board assembly.

[0040] As shown in FIG. 3, in an example provided in the embodiments, the power board assembly **10** includes a power board **11** and a regulation board **12**. The regulation board **12** is disposed on a surface of the power board **11**, and the regulation board **12** is perpendicular to the power board **11**. The power board **11** includes a conductive wire **111** and a switching component **112**. The switching component **112** is connected to the conductive wire **111** in series, and is configured to connect or disconnect the conductive wire **111**. The regulation board **12** includes a semi-closed Rogowski coil **121** and a regulation circuit **122**, and the regulation circuit **122** is connected to the semi-closed Rogowski coil **121**. The semi-closed Rogowski coil **121** crosses the conductive wire **111**, and the semi-closed Rogowski coil **121** is configured to: induce a current in the conductive wire **111** and generate an induction signal, to effectively detect the current in the conductive wire **111**. The regulation circuit **122** is configured to generate a detection signal after performing operation processing on the induction signal. The power board assembly **10** further includes a controller **113**. The controller **113** is disposed on the power board **11**. The controller **113** is connected to the regulation circuit **122** and the switching component **112**. The controller **113** is configured to control on or off of the switching component **112** based on the detection signal of the regulation circuit **122**, to implement a function like current short-circuit protection.

[0041] For example, in the example provided in the embodiments, the regulation board **12** is perpendicularly disposed on a surface of the power board **11**, so that a board area occupied by the regulation board **12** on the power board **11** can be effectively reduced. This helps implement a miniaturization design of the power board **11**. In addition, more other electronic components can be disposed on the power board **11**, to improve functions of the power board **11**. After the semi-closed Rogowski coil **121** is used in the regulation board **12**, the regulation board **12** can be disposed on the surface of the power board **11** in a form of single-side layout, and a good current detection function can be implemented. After the regulation board **12** is disposed on the surface of the power board **11**, the semi-closed Rogowski coil **121** crosses the conductive wire **111** in the power board **11**. When an alternating current flows through the conductive wire **111**, the semi-closed Rogowski coil **121** may generate the corresponding induction signal according to an electromagnetic induction principle. The regulation circuit **122** generates the corresponding detection signal after performing operation processing on the induction signal. The controller **113** may control on or off of the switching component **112** based on the detection signal, to ensure use safety of the power board **11**.

[0042] Thus, in the example provided in the embodiments, the semi-closed Rogowski coil **121** has an advantage of low costs, which is conducive to wide application. The semi-closed Rogowski coil **121** is used, so that the regulation board **12** can be perpendicularly disposed on a side of the power board **11**, and a good current detection function is implemented. In addition, the regulation board

12 occupies a small board area in the power board **11**, which helps implement a miniaturization design of the power board assembly **10**, so that the power board assembly **10** can be better used in a device having a high requirement on space. The regulation board **12** being perpendicular to the power board **11** means that the regulation board **12** is approximately perpendicular to the power board **11**, that is, an included angle between the regulation board **12** and the power board **11** may be about 90 degrees, which is not perpendicular in a strict sense.

[0043] The controller **113** being configured to control on or off of the switching component **112** based on the detection signal of the regulation circuit **122** means that the controller **113** further needs to refer to the detection signal when switching an on state or an off state of the switching component **112** based on the control signal. Alternatively, it may be understood that, when the detection signal meets a condition, the controller **113** may switch the on state or the off state of the switching component **112**. When the detection signal does not meet the condition, the controller **113** cannot switch the on state or the off state of the switching component **112**.

[0044] For example, when a current in the conductive wire **111** is less than a preset value, the regulation circuit **122** generates a corresponding detection signal. In this case, the detection signal meets the condition, and the controller **113** may switch the on state or the off state of the switching component **112**.

[0045] Alternatively, when the current in the conductive wire **111** is greater than or equal to the preset value, the regulation circuit **122** generates a corresponding detection signal. In this case, the detection signal does not meet the condition, and the controller **113** cannot switch the on state or the off state of the switching component **112**.

[0046] Alternatively, it may be understood that, in some cases, the switching component **112** does not have a turn-on/off capability under a large current; otherwise, a bad case like arcing may occur. Therefore, the switching component **112** is allowed to switch between the on state and the off state only when the current in the conductive wire **111** is small, so that use safety of the switching component **112** can be ensured.

[0047] It should be noted that a Rogowski coil is usually a hollow ring coil, and may be directly sleeved on the measured conductive wire **111**. An alternating current flowing through the conductive wire **111** generates an alternating magnetic field around the conductive wire **111**, so that an alternating-current voltage signal that is proportional to a current change ratio is induced in the Rogowski coil. An alternating-current voltage signal is obtained after the voltage signal output by the Rogowski coil is integrated by using an integrator or an integral circuit. The alternating-current voltage signal can accurately reproduce a waveform of a measured current signal.

[0048] A working principle of the semi-closed Rogowski coil **121** is substantially the same as that of the conventional Rogowski coil. In the example provided in the embodiments, the semi-closed Rogowski coil **121** may be considered as a structural variation of the Rogowski coil. That is, the Rogowski coil is in an approximately closed ring shape, and the semi-closed Rogowski coil **121** is in an approximately U or semi-ring shape. The semi-closed Rogowski coil **121** is configured to: induce a current in the conductive wire **111** and generate an induction signal. The induction signal is a voltage signal. The regulation circuit **122** is configured to generate a detection signal after performing operation processing on the induction signal (such as the voltage signal).

[0049] In the example provided in the embodiments, the regulation circuit **122** includes an integrator and an amplifier. The integrator is configured to perform integration on the induction signal (such as the voltage signal) generated by the semi-closed Rogowski coil **121** to obtain an alternating current voltage signal, and the amplifier amplifies the alternating current voltage signal and outputs the amplified alternating current voltage signal. That is, the detection signal generated by the foregoing regulation circuit **122** is the alternating current voltage signal processed by the integrator and the amplifier. The controller **113** controls, based on the alternating current voltage signal, the switching component **112** to be disconnected or connected.

[0050] During actual application, there may be various specific types of the regulation circuit **122**.

For example, as shown in FIG. 4, an embodiment provides a circuit diagram of the regulation circuit **122**. A left end of the regulation circuit **122** is used as an input end, and is configured to connect to the semi-closed Rogowski coil **121** to receive the induction signal of the semi-closed Rogowski coil **121**. A right end of the regulation circuit **122** is used as an output end, and is configured to connect to the controller **113** to output the detection signal.

[0051] It should be noted that, during specific setting, the regulation circuit **122** may use a currently commonly used type and circuit structure. This is not limited.

[0052] In the example provided above, the controller **113** is disposed on the surface of the power board **11**. In another example, the controller **113** may alternatively be disposed on a surface of the regulation board **12**. A disposing position of the controller **113** is not limited.

[0053] As shown in FIG. 3, during specific application, the power board **11** may include a printed circuit board. For example, the power board **11** includes a dielectric substrate **110**, and a copper layer is provided on a surface of the dielectric substrate **110**. The copper layer may form the conductive wire **111**. Electronic components such as the controller **113** and the switching component **112** may be disposed on a surface of the dielectric substrate **110** and connected to the conductive wire **111** (not shown in the figure) in the dielectric substrate **110** through a surface-mount technology or the like. During actual application, parameters such as a width, a thickness, and a form of the conductive wire **111** may be appropriately set based on an actual requirement. Details are not described herein.

[0054] In addition, the regulation board **12** may also include a printed circuit board. For example, the regulation board **12** includes a dielectric substrate **120**, and components such as an inductor and a capacitor in the regulation circuit **122** may be disposed on a surface of the dielectric substrate **120** and connected to a conductive wire **111** (not shown in the figure) in the dielectric substrate **120** through a surface-mount technology or the like.

[0055] The power board **11** and the regulation board **12** may alternatively be disposed in various manners.

[0056] For example, as shown in FIG. 5, in an example provided in the embodiments, an edge of the regulation board **12** includes two conductive protrusions **123**, and the regulation circuit **122** is electrically connected to the conductive protrusions **123**. The power board **11** includes two conductive holes **114**, and the two conductive protrusions **123** are respectively inserted into the two conductive holes **114**, to implement conductive connections between the conductive protrusions **123** and the conductive holes **114**. In summary, a signal transmission function may be implemented between the regulation board **12** and the power board **11** through conductive connections between the conductive holes **114** and the conductive protrusions **123**. The detection signal generated by the regulation circuit **122** in the regulation board **12** may be transmitted to the controller **113** in the power board **11** through conductive paths between the conductive holes **114** and the conductive protrusions **123**. That is, in the embodiments, through insertion connections between the conductive protrusions **123** and the conductive holes **114**, fastening between the regulation board **12** and the power board **11** can be implemented, and a signal connection between the regulation circuit **122** and the controller **113** can be further implemented.

[0057] In addition, in the example provided in the embodiments, the regulation board **12** includes the two conductive protrusions **123**, and both the two conductive protrusions **123** are located at an edge of a same side of the regulation board **12**. The power board **11** includes the two conductive holes **114**. After the two conductive protrusions **123** are inserted into the two conductive holes **114**, relative rotation between the regulation board **12** and the power board **11** can be effectively prevented, thereby effectively improving connection stability and connection strength between the regulation board **12** and the power board **11**.

[0058] As shown in FIG. 5, the two conductive holes **114** are located on two sides of the conductive wire **111**. After the two conductive protrusions **123** are inserted into the two conductive holes **114**, the two conductive protrusions **123** are also located on the two sides of the conductive

wire **111**, and the semi-closed Rogowski coil **121** crosses the conductive wire **111**.

[0059] During specific setting, the semi-closed Rogowski coil **121** may be in an approximately semi-circular ring shape. After the regulation board **12** is inserted into the power board **11**, the conductive wire **111** may be approximately located in a center of the semi-closed Rogowski coil **121**, thereby ensuring detection effect of the semi-closed Rogowski coil **121**. The two conductive holes **114** may be symmetrically disposed with respect to the conductive wire **111**, and the two conductive protrusions **123** may be respectively located at two ends of the semi-closed Rogowski coil **121**, so that after the regulation board **12** is inserted into the power board **11**, the conductive wire **111** can be located at the center of the semi-closed Rogowski coil **121**.

[0060] In addition, as shown in FIG. **6**, in the example provided in the embodiments, the regulation board **12** further has a notch **124**, and the notch **124** directly faces the conductive wire **111**. After the regulation board **12** is mounted on the power board **11**, the notch **124** is located right above the conductive wire **111**. Disposing the notch **124** can reduce a weight of the regulation board **12**, and help implement heat dissipation of the conductive wire **111**. For example, the notch **124** is disposed, so that a channel for air circulation may be formed between the conductive wire **111** and the regulation board **12**, and when air flows through the notch **124**, heat generated by the conductive wire **111** may be taken away, thereby ensuring heat dissipation performance of the conductive wire **111**.

[0061] During specific setting, a form and a size of the notch **124** may be appropriately set based on an actual requirement. This is not limited.

[0062] It may be understood that, during actual application, the conductive protrusion **123** may be a tin connector. After the conductive protrusion **123** is inserted into the conductive hole **114**, a reliable connection between the conductive protrusion **123** and the conductive hole **114** may be implemented in a soldering manner. In addition, the conductive hole **114** may be a plated hole, or may be a structure like a through hole or a blind hole with a conductive material on an inner wall. During specific disposing, specific structures and forms of the conductive hole **114** and the conductive protrusion **123** may be flexibly adjusted based on an actual requirement. This is not limited.

[0063] In another example, the regulation board **12** may alternatively include one conductive protrusion **123**, and the power board **11** may alternatively include one conductive hole **114**. A signal connection function between the regulation board **12** and the power board **11** may be implemented through an insertion connection between the conductive hole **114** and the conductive protrusion **123**.

[0064] Alternatively, in some examples, the regulation board **12** may alternatively include two protrusions, and the power board **11** may alternatively include two through holes. One of the protrusions may be a protrusion having a conductive function, and the other protrusion may not have the conductive function. Correspondingly, one through hole may be a through hole having a conductive function, and the other through hole may not have the conductive function. That is, after the protrusion with the conductive function is inserted into the through hole with the conductive function, a mechanical connection and a conductive connection between the power board **11** and the regulation board **12** may be implemented. After the protrusion without the conductive function is inserted into the through hole without the conductive function, the mechanical connection between the power board **11** and the regulation board **12** may be implemented.

[0065] In another example, the regulation board **12** may alternatively include three or more protrusions, and the power board **11** may alternatively include three or more through holes. During actual application, quantities and location layouts of the protrusions and the through holes may be appropriately set based on an actual requirement. Details are not described herein.

[0066] When the semi-closed Rogowski coil **121** is disposed, the semi-closed Rogowski coil **121** may also be disposed in various manners.

[0067] For example, as shown in FIG. **6**, in an example provided in the embodiments, the semi-

closed Rogowski coil **121** is located in the dielectric substrate, so that the dielectric substrate can effectively protect the semi-closed Rogowski coil **121**, and prevent an external force from directly acting on the semi-closed Rogowski coil **121**. Therefore, the semi-closed coil can be effectively protected, and security and reliability of the semi-closed Rogowski coil **121** can be provided.

[0068] Alternatively, in another example, the semi-closed Rogowski coil **121** may alternatively be disposed on a side of the dielectric substrate. Alternatively, it may be understood that, the semi-closed Rogowski coil **121** may be disposed on the surface of the dielectric substrate, effectively facilitating preparation of the semi-closed Rogowski coil **121**.

[0069] Alternatively, in some examples, after the semi-closed Rogowski coil **121** is disposed on a side of the dielectric substrate, the semi-closed Rogowski coil **121** may be further packaged through a packaging process, so that a packaging material like resin effectively wraps the semi-closed Rogowski coil **121**, thereby improving security and reliability of the semi-closed Rogowski coil **121**.

[0070] During specific disposing, a disposing manner of the semi-closed Rogowski coil **121** may be properly selected and adjusted based on an actual requirement. Details are not described herein.

[0071] In addition, as shown in FIG. **6**, in an example provided in the embodiments, the semi-closed Rogowski coil **121** is located between the regulation circuit **122** and the conductive wire **111**. In other words, the regulation circuit **122** is located on a side that is of the semi-closed Rogowski coil **121** and that is away from the conductive wire **111**, so that the regulation circuit **122** does not cause interference to normal operation of the semi-closed Rogowski coil **121**. In addition, the semi-closed Rogowski coil **121** is located between the regulation circuit **122** and the conductive wire **111**, so that the semi-closed Rogowski coil **121** can form an effective electromagnetic shielding effect, and can prevent an alternating magnetic field generated near the conductive wire **111** from interfering with the regulation circuit **122**.

[0072] It may be understood that, during specific setting, relative positions of the regulation circuit **122** and the semi-closed Rogowski coil **121** may be appropriately set based on an actual requirement, to ensure stable operation of the regulation circuit **122** and the semi-closed Rogowski coil **121**. Details are not described herein.

[0073] In addition, as shown in FIG. **7**, in the example provided in the embodiments, both output ends of the semi-closed Rogowski coil **121** are located on a same side of the semi-closed Rogowski coil **121**, to connect to the regulation circuit **122**.

[0074] For example, the semi-closed Rogowski coil **121** includes a first end **121a** (for example, a left end in FIG. **7**) and a second end **121b** (for example, a right end in FIG. **7**). The semi-closed Rogowski coil **121** includes a spiral wire **1210**. Both ends of the spiral wire **1210** are used as output ends and are located at the first end **121a**.

[0075] In the example provided in the embodiments, in the semi-closed Rogowski coil **121**, one end of the spiral wire **1210** is located at the first end **121a**, and the other end of the spiral wire **1210** extends from the second end **121b** along an axis of the semi-closed Rogowski coil **121** to the first end **121a**. It should be noted that the axis of the semi-closed Rogowski coil **121** is a virtual axis of the semi-closed Rogowski coil **121**. In the example provided in the embodiments, the semi-closed Rogowski coil **121** is in an approximately U form as a whole, in other words, the axis of the semi-closed Rogowski coil **121** is in an approximately U form. In other words, when the semi-closed Rogowski coil **121** is prepared, the spiral wire **1210** is wound along the axis. After the winding is completed, one end of the spiral wire **1210** extends in a path of the axis to the other end, so that the two ends of the spiral wire **1210** are located at the same end of the entire semi-closed Rogowski coil **121**. In this disposing manner, internal space of the semi-closed Rogowski coil **121** can be appropriately used, and the semi-closed Rogowski coil **121** can be easily connected to the regulation circuit **122**.

[0076] It may be understood that, during actual application, a parameter of the semi-closed Rogowski coil **121** may be appropriately set based on an actual requirement. This is not limited.

The parameter of the semi-closed Rogowski coil **121** includes, but is not limited to, a form, a quantity of circles, a diameter of the spiral wire, and the like of the semi-closed Rogowski coil **121**. [0077] In addition, in the foregoing example, the switching component **112** may be any one of a relay, an insulated gate bipolar transistor, or a metal-oxide-semiconductor field-effect transistor. A specific type of the switching component **112** is not limited.

[0078] During actual application, the power board assembly **10** may be used in a plurality of devices or scenarios in which a current connection and disconnection is implemented.

[0079] For example, as shown in FIG. **8**, an embodiment further provides a backup power device **20**. The backup power device **20** may also be referred to as a backup power box, a backup power case, or the like. The backup power device **20** may include a housing **21**, an input port **22**, an output port **23**, and the foregoing power board assembly **10**. Both the input port **22** and the output port **23** penetrate the housing **21**, to implement connection between the backup power device **20** and another external power supply device or an external power-consuming device. The power board assembly **10** is located in the housing **21**, and the conductive wire **111** is connected between the input port **22** and the output port **23**. An external alternating current may be transmitted from the input port **22** to the backup power device **20**, and then output from the output port **23**. In the backup power device **20** provided in embodiments, a current detection function may be implemented through the foregoing power board assembly **10**, and use safety and reliability are good. In addition, the power board assembly **10** has advantages of miniaturization and low costs, so that the backup power device **20** using the power board assembly **10** also has advantages of miniaturization and low costs, which facilitates wide application of the backup power device **20**, and the backup power device **20** can be well used in a scenario having a high requirement on space.

[0080] In addition, as shown in FIG. **8**, in an example provided in the embodiments, the backup power device **20** may further include a circuit breaker **24**. The circuit breaker **24** is connected between the conductive wire **111** and the output port of the power board **11**, and is configured to connect or disconnect a conductive path between the conductive wire **111** and the output port. The circuit breaker **24** may be a device having a large current connection or disconnection capability, for example, an air switch. It may be understood that, in the power board assembly **10**, to ensure safety of the power board assembly **10**, the switching component **112** usually does not have a capability of connecting and disconnecting a large current. Therefore, in the example provided in the embodiments, an additional circuit breaker **24** is disposed in the backup power device **20**, so that when a working condition like a short circuit occurs on an external power-consuming device connected to the output port **23**, a transmission path of a current may be cut off by using the circuit breaker **24**. In this case, the switching component **112** in the power board assembly **10** may keep in an on state.

[0081] For example, when a current in the conductive wire **111** is greater than or equal to a preset value, the regulation circuit generates a corresponding detection signal. In this case, the controller **113** cannot switch the on state or an off state of the switching component **112**. Therefore, the controller **113** may cut off the transmission path of the current by controlling the circuit breaker **24**.

[0082] During actual application, the backup power device **20** may be used in a plurality of scenarios.

[0083] For example, as shown in FIG. **9**, an embodiment further provides a photovoltaic system, including a photovoltaic power generation device, an inverter, and a backup power device. The photovoltaic power generation device is configured to convert light energy into a direct current, and the inverter is configured to: convert the direct current generated by the photovoltaic power generation device into an alternating current and then transmit the alternating current to a power grid. The backup power device is connected between the inverter and the power grid, and is configured to implement connection or disconnection between the inverter and the power grid. In other words, the backup power device may connect a power supply path between the inverter and the power grid, so that electric energy generated by the photovoltaic power generation device can

be transmitted to the power grid. Alternatively, the backup power device may disconnect the power supply path between the inverter and the power grid, so that the electric energy generated by the photovoltaic power generation device cannot be transmitted to the power grid.

[0084] A power-consuming device like a home load may be connected to the power grid, to ensure normal use of the power-consuming device. During actual use, a working condition like a short circuit may occur in the power-consuming device, resulting in a large current in a conductive wire 111 in the backup power device. In this case, a circuit breaker of the power-consuming device may be used to cut off a transmission path of the current. Alternatively, the current transmission path between the backup power device and the power grid may be cut off by using a circuit breaker in the backup power device.

[0085] It may be understood that, during actual application, when a working condition like a short circuit occurs in the power-consuming device, a on or off state of the circuit breaker in the backup power device may be flexibly switched based on an actual requirement. This is not limited.

[0086] In various embodiments, unless otherwise stated or there is a logic conflict, terms and/or descriptions in different embodiments are consistent and may be mutually referenced, and features in different embodiments may be combined based on an internal logical relationship thereof, to form a new embodiment.

[0087] “A plurality of” in the embodiments refers to two or more than two. “And/or” describes an association relationship between associated objects and indicates that three relationships may exist. For example, A and/or B may indicate the following three cases: only A exists, both A and B exist, and only B exists, where A and B may be singular or plural.

[0088] It may be understood that various numbers in embodiments are merely used for differentiation for ease of description and are not used to limit the scope of embodiments. Sequence numbers of the foregoing processes do not mean an execution sequence, and the execution sequence of the processes should be determined based on functions and internal logic of the processes.

Claims

1. A backup power device, comprising: a power board comprising: a conductive wire, a switching component connected to the conductive wire in series, and a controller; and a regulation board comprising: a semi-closed Rogowski coil, and a regulation circuit connected to the semi-closed Rogowski coil, wherein the regulation board is disposed on a surface of the power board, and the regulation board is perpendicular to the power board, the semi-closed Rogowski coil crosses the conductive wire, and is configured to: induce a current in the conductive wire and generate an induction signal, and the regulation circuit is configured to: generate a detection signal after performing operation processing on the induction signal; and the controller is configured to control on or off of the switching component based on the detection signal of the regulation circuit.
2. The backup power device according to claim 1, wherein the controller is further configured to: when the current in the conductive wire is less than a preset value, control on or off of the switching component.
3. The backup power device according to claim 1, wherein the controller is further configured to: when the current in the conductive wire is greater than or equal to the preset value, control the switching component to remain on or off.
4. The backup power device according to claim 1, wherein the semi-closed Rogowski coil is in a semi-circular ring shape, and the conductive wire penetrates a center of the semi-closed Rogowski coil.
5. The backup power device according to claim 1, wherein the regulation board further comprises: a dielectric substrate, and the semi-closed Rogowski coil is located in the dielectric substrate.
6. The backup power device according to claim 1, wherein the regulation board further comprises:

a dielectric substrate, and the semi-closed Rogowski coil is located on a side of the dielectric substrate.

7. The backup power device according to claim 1, wherein the semi-closed Rogowski coil is located between the regulation circuit and the conductive wire.

8. The backup power device according to claim 1, wherein the semi-closed Rogowski coil comprises a first end and a second end, the semi-closed Rogowski coil is formed by a spiral wire, and both ends of the spiral wire are located at the first end.

9. The backup power device according to claim 8, wherein, in the semi-closed Rogowski coil, a first end of the spiral wire is located at the first end and a second end of the spiral wire penetrates the second end along an axis of the semi-closed Rogowski coil to the first end.

10. The backup power device according to claim 1, wherein an edge of the regulation board further comprises: a conductive protrusion, the regulation circuit is electrically connected to the conductive protrusion.

11. The backup power device according to claim 10, wherein the regulation board further comprises: two conductive protrusions that are both located at an edge of a same side of the regulation board.

12. The backup power device according to claim 1, wherein the switching component is any one of a relay, an insulated gate bipolar transistor, and a metal-oxide-semiconductor field-effect transistor.

13. The backup power device according to claim 1, further comprising: a housing, an input port, and an output port, both the input port and the output port penetrate the housing, both the power board and the regulation board are located in the housing, and the conductive wire is connected between the input port and the output port.

14. The backup power device according to claim 1, further comprising: a circuit breaker connected between the conductive wire and the output port, and configured to connect or disconnect a conductive path between the conductive wire and the output port.

15. The backup power device according to claim 14, wherein the controller is further connected to the circuit breaker, and further configured to: when the current in the conductive wire is greater than or equal to the preset value, control the circuit breaker to be disconnected.

16. A photovoltaic system comprising a photovoltaic power generation device; an inverter; and a backup power device, wherein the inverter is connected between the photovoltaic power generation device and the backup power device and the backup power device comprises: a power board comprising: a conductive wire, a switching component connected to the conductive wire in series, and a controller; and a regulation board comprising: a semi-closed Rogowski coil, and a regulation circuit connected to the semi-closed Rogowski coil, wherein the regulation board is disposed on a surface of the power board, and the regulation board is perpendicular to the power board, the semi-closed Rogowski coil crosses the conductive wire, and is configured to: induce a current in the conductive wire and generate an induction signal, and the regulation circuit is configured to: generate a detection signal after performing operation processing on the induction signal; and the controller is configured to control on or off of the switching component based on the detection signal of the regulation circuit; the inverter is connected to the input port; and the inverter is configured to: convert a direct current generated by the photovoltaic power generation device into an alternating current, and then transmit the alternating current to the input port, wherein the output port is configured to connect to a power grid.

17. The backup power device according to claim 1, wherein the semi-closed Rogowski coil comprises a first end and a second end, the semi-closed Rogowski coil is formed by a spiral wire, and both ends of the spiral wire are located at second end.

18. The backup power device according to claim 8, wherein, in the semi-closed Rogowski coil, a first end of the spiral wire is located at the second end, and a second end of the spiral wire penetrates the first end along an axis of the semi-closed Rogowski coil to the second end.

19. The backup power device according to claim 10, wherein the power board further comprises: a

conductive hole, the conductive protrusion is inserted into the conductive hole, and the conductive protrusion is electrically connected to the conductive hole.

20. The backup power device according to claim 11, wherein the power board further comprises: two conductive holes connected to the two conductive protrusions in a one-to-one correspondence and the two conductive holes are distributed on two sides of the conductive wire.
